

Hierarchical Adaptive Isogeometric Analysis for J_2 Plasticity:

A Computational Study

Project report

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Abstract

This report summarizes the analyses carried out in the development of an adaptive isogeometric solver for J_2 plasticity based on truncated hierarchical B-splines (THB-splines). We document the mathematical framework, the projection strategies for history variables (global L^2 , local quasi-interpolation in C^{p-1} and in C^0 with Tikhonov regularization), and two element-local error estimators (residual-type $\|\mathbf{f} + \nabla \cdot \boldsymbol{\sigma}_h\|$ and gradient-seminorm $\|\nabla \boldsymbol{\sigma}_h\|_F$). We present three numerical studies on the eighth-sphere benchmark in 3D and on the plane-strain ring in 2D, both under monotonically increasing internal pressure, and compare against Hill's analytical solution. We discuss the implementation of the estimators, including a bug fix and a $\mathcal{O}(10^2\text{--}10^3)\times$ performance optimization of the divergence estimator, and we conclude with an honest assessment of when local quasi-interpolation can be expected to outperform a global L^2 projection in this context.

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1 Introduction

Adaptive mesh refinement is a classical technique to concentrate computational effort where the local error is large. In the context of isogeometric analysis (IGA) with hierarchical splines, refinement operates on a sequence of nested tensor-product B-spline spaces and relies on two ingredients: *(i)* an a-posteriori error indicator η_K for each active element K , and *(ii)* a projection operator that transfers history variables (plastic strain $\boldsymbol{\varepsilon}^p$, accumulated plastic strain, stress $\boldsymbol{\sigma}$) between meshes when the active set changes.

In small-strain J_2 plasticity, both ingredients have to cope with the sharp *elastic-plastic front*: the moving interface separating the elastic core from the plasticised region, across which the stress gradient is large but bounded. Resolving the front efficiently is the practical motivation for hierarchical refinement; transferring the plastic state across mesh changes without spurious oscillations is the practical motivation for studying alternative projection strategies.

This report covers:

- the mathematical framework (§2);
- the three projection strategies considered — global L^2 , local quasi-interpolation $\text{QI}_{C^{p-1}}$, and Tikhonov-regularized local quasi-interpolation QI_{C^0} (§3);
- two element-local error estimators (§4);
- three numerical studies: a 3D adaptivity sweep over $\max_lvl \in \{1, 2, 3, 4\}$, a 2D analogue, and an estimator comparison (§5);
- a critical discussion of the observed performance (§6).

The implementation extends the `geopdes_hierarchical_plasticity` package, builds on the GeoPDEs library (v3.2.2) and the NURBS toolbox (v1.4.3), and runs on MATLAB R2024b.

2 Mathematical framework

2.1 Strong-form problem and constitutive model

We consider quasi-static small-strain elasto-plasticity on a domain $\Omega \subset \mathbb{R}^d$ ($d \in \{2, 3\}$). Find the displacement $\mathbf{u} : \Omega \rightarrow \mathbb{R}^d$ such that

$$-\nabla \cdot \boldsymbol{\sigma}(\mathbf{u}) = \mathbf{f} \quad \text{in } \Omega, \quad (1)$$

$$\mathbf{u} = \mathbf{0} \quad \text{on } \Gamma_D, \quad (2)$$

$$\boldsymbol{\sigma}(\mathbf{u}) \cdot \mathbf{n} = \mathbf{t} \quad \text{on } \Gamma_N, \quad (3)$$

where $\boldsymbol{\sigma}$ is the Cauchy stress, $\mathbf{f}$ is the body force (set to $\mathbf{0}$ in our benchmarks), and $\Gamma_D \cup \Gamma_N = \partial\Omega$. The strain decomposes additively as

$$\boldsymbol{\varepsilon}(\mathbf{u}) = \frac{1}{2}(\nabla \mathbf{u} + \nabla \mathbf{u}^\top) = \boldsymbol{\varepsilon}^e + \boldsymbol{\varepsilon}^p, \quad (4)$$

and the constitutive law is the standard J_2 flow rule with perfect plasticity:

$$\boldsymbol{\sigma} = \mathbb{C} : \boldsymbol{\varepsilon}^e = \mathbb{C} : (\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^p), \quad (5)$$

$$\dot{\boldsymbol{\varepsilon}}^p = \dot{\gamma} \frac{\partial f}{\partial \boldsymbol{\sigma}}, \quad f(\boldsymbol{\sigma}) = \sqrt{\frac{3}{2} \text{dev} \boldsymbol{\sigma} : \text{dev} \boldsymbol{\sigma}} - \sigma_y, \quad (6)$$

with $\mathbb{C}$ the linear elastic stiffness tensor (Lamé parameters $\lambda_{\text{Lame}}, \mu_{\text{Lame}}$), $\text{dev} \boldsymbol{\sigma} = \boldsymbol{\sigma} - \frac{1}{3} \text{tr}(\boldsymbol{\sigma}) \mathbf{I}$ the deviatoric stress, σ_y the yield stress, $\dot{\gamma} \geq 0$ the plastic multiplier, and $f \leq 0, \dot{\gamma} f = 0$ the Karush-Kuhn-Tucker conditions.

The discrete problem at each load step uses a return-mapping algorithm (radial return) inside a Newton iteration on the global residual.

2.2 Hill's analytical reference

For both benchmarks the analytical reference is given by Hill's solution for an internally pressurised hollow sphere/cylinder under elastic-perfectly-plastic behaviour [1]. Given inner radius a , outer radius b , internal pressure P , Young's modulus E and Poisson's ratio ν , the plastic radius c satisfies

$$P = \frac{2}{3} \sigma_y [3 \ln(c/a) + 1 - (c/b)^3] \quad (\text{sphere case}), \quad (7)$$

and a closed-form expression provides the radial and tangential stresses $\sigma_r(r), \sigma_\theta(r)$ in both elastic and plastic regions, as well as the radial displacement u_r at the outer boundary. We use this as ground truth in all error metrics.

2.3 Hierarchical THB-splines

Let $\{V^\ell\}_{\ell=0}^{N-1}$ be a sequence of nested tensor-product B-spline spaces of degree p on Ω , obtained by dyadic refinement: $V^\ell \subset V^{\ell+1}$. Let Ω^ℓ be the *active* subdomain at level ℓ , with $\Omega^{\ell+1} \subset \Omega^\ell$. The hierarchical mesh consists of the active cells at each level, and the truncated hierarchical (THB) basis [2] is

$$\mathcal{H} = \bigcup_{\ell=0}^{N-1} \left\{ \text{trunc}^{\ell+1}(B_i^\ell) : \text{supp } B_i^\ell \subset \Omega^\ell, \text{supp } B_i^\ell \not\subset \Omega^{\ell+1} \right\}, \quad (8)$$

where $\text{trunc}^{\ell+1}$ removes the components in $V^{\ell+1}$. The truncation gives the basis the partition-of-unity property and limits the support overlap between coarse and fine functions, which is beneficial for both the linear algebra and the design of projection operators. All hierarchical operations (refine, coarsen, mark) are provided by GeoPDEs [3].

3 Projection of history variables

Each load step needs a representation of the plastic strain $\boldsymbol{\varepsilon}^p$ and stress $\boldsymbol{\sigma}$ on the current hierarchical scalar space $\mathcal{H}_\sigma$. Three strategies are implemented and compared.

3.1 Global L^2 projection

The classical option: given quadrature-point values $\boldsymbol{\sigma}$ defined on the displacement mesh, find the unique $\boldsymbol{\sigma}_h \in (\mathcal{H}_\sigma)^6$ (Voigt) such that

$$\int_{\Omega} \boldsymbol{\sigma}_h \cdot \boldsymbol{\varphi} \, d\Omega = \int_{\Omega} \boldsymbol{\sigma} \cdot \boldsymbol{\varphi} \, d\Omega \quad \forall \boldsymbol{\varphi} \in (\mathcal{H}_\sigma)^6, \quad (9)$$

which assembles to a sparse hierarchical mass matrix M and a per-Voigt right-hand side $b_c = \int_{\Omega} \boldsymbol{\sigma}_c \boldsymbol{\varphi}$. The projected coefficients solve $Mx_c = b_c$. Optimal in the L^2 norm; requires a global solve at every projection step.

3.2 Local quasi-interpolation $\text{QI}_{C^{p-1}}$

A local alternative: for each active hierarchical basis function B_k with support Ω_k , fit a B-spline expansion σ_h on a local patch of cells around Ω_k in the (single-level) tensor-product sense, by penalised least squares in the data points $\{\mathbf{x}_q\}$ (Gauss quadrature points lying in the patch):

$$\min_{\hat{\sigma}} \sum_q w_q |\hat{\sigma}(\mathbf{x}_q) - \sigma(\mathbf{x}_q)|^2 + \lambda \hat{\sigma}^\top M_{\nabla^2} \hat{\sigma}, \quad (10)$$

where w_q are quadrature weights and M_{∇^2} is the thin-plate (gradgrad) Gramian. $\hat{\sigma}(B_k \text{ coefficient})$ is then assigned to the corresponding hierarchical degree of freedom. In our $\text{QI}_{C^{p-1}}$ variant the projection space inherits the maximum regularity (C^{p-1}) of the underlying B-spline space and uses $\lambda = 0$ (no regularization needed: the local LS systems are well conditioned). The MATLAB implementation uses a C/OpenMP MEX backend (`libqi`) so the per-DOF loop runs in compiled code.

3.3 Local quasi-interpolation QI_{C^0}

Same procedure as $\text{QI}_{C^{p-1}}$, but the projection space is re-instantiated with internal knot multiplicity equal to p , **forcing C^0 regularity**. This is the line in the solver

```
if strcmpi(method_data.type_projection, 'QI_C0')
    regularity_projection = degree_projection - degree_projection; % =
    0
end
```

With C^0 basis, the basis-function supports are smaller and the local LS systems become more ill-conditioned. We therefore enable **Tikhonov regularization** by setting $\lambda = 10^{-9}$ in Eq. (10). The penalty term shifts every eigenvalue of the local normal-equations matrix away from zero by at least λ times the smallest eigenvalue of M_{∇^2} , and biases the solution toward smoothness of the second derivative; the value 10^{-9} is small enough to barely perturb well-conditioned patches and large enough to rescue ill-conditioned ones.

Why C^0 ? In a plasticity setting the projected stress is piecewise smooth with sharp gradients near the elastic-plastic front. A C^0 representation can capture jumps in the derivative at every knot line, which is exactly what is happening at the front (the stress is continuous but has a kink), whereas a C^{p-1} basis enforces unwanted smoothness across the front and produces visible Gibbs-like overshoots.

4 Element-local error estimators

We implement and compare two element-local indicators, both producing a vector η_K for each active element K , used by the Dörfler/maximum-strategy marker.

4.1 Residual-type, divergence of stress

Following the classical residual a-posteriori theory [4],

$$\eta_K^{\text{div}} = h_K \|\mathbf{f} + \nabla \cdot \sigma_h\|_{L^2(K)}, \quad (11)$$

where h_K is the element diameter. The estimator is reliable up to a constant for linear elasticity; in plasticity it remains a sound proxy since the constitutive law enters only through σ_h .

Implementation note — bug. The original 3D implementation contained a structural bug in the Voigt-to-divergence assembly: the σ_{zz} , σ_{yz} and σ_{xz} contributions were nested *inside* the `elseif ifield == 4` branch and therefore never executed. The divergence in 3D was missing $\partial\sigma_{xz}/\partial z$, $\partial\sigma_{yz}/\partial z$, and the entire third row. We refactored the contribution accumulation into an explicit `switch/case` dispatch driven by the Voigt index, which fixes the bug and makes the dimension dependence explicit:

```
function divergence = accumulate_divergence(divergence, hgrad, ifield,
    ncomp)
    switch ifield
        case 1 %
            sigma_xx
            divergence(1,:,:) = divergence(1,:,:) + hgrad(1,:,:);
        case 2 %
            sigma_yy
            divergence(2,:,:) = divergence(2,:,:) + hgrad(2,:,:);
        case 3 %
            sigma_zz (3D only)
            if ncomp == 3, divergence(3,:,:) = divergence(3,:,:) + hgrad
                (3,:,:); end
        case 4 %
            sigma_xy
            divergence(1,:,:) = divergence(1,:,:) + hgrad(2,:,:);
            divergence(2,:,:) = divergence(2,:,:) + hgrad(1,:,:);
        case 5 %
            sigma_yz (3D only)
            if ncomp == 3
                divergence(2,:,:) = divergence(2,:,:) + hgrad(3,:,:);
                divergence(3,:,:) = divergence(3,:,:) + hgrad(2,:,:);
            end
        case 6 %
            sigma_xz (3D only)
            if ncomp == 3
                divergence(1,:,:) = divergence(1,:,:) + hgrad(3,:,:);
                divergence(3,:,:) = divergence(3,:,:) + hgrad(1,:,:);
            end
    end
end
```

Implementation note — performance. The original implementation evaluated $\nabla\sigma$ point by point through `sp_eval_phys`, which performs a Newton inversion of the NURBS map per quadrature point. With n_{el} elements, n_{qn} quadrature nodes per element and 6 Voigt components, the total number of Newton solves is $6n_{\text{el}}n_{\text{qn}} \approx 24,000$ for our 3D meshes, taking minutes per ESTIMATE call. We replaced the triple-nested loop with 6 *bulk* `hspace_eval_hmsh` calls — one per Voigt component — which exploit the pre-existing hierarchical mesh quadrature structure with no inverse mapping. The legacy path is retained behind a flag for the case where $\mathcal{H}_\sigma$ lives on a refined projection mesh. The expected speed-up is two to three orders of magnitude.

4.2 Stress-gradient seminorm

A heuristic but robust alternative:

$$\eta_K^{\nabla\sigma} = h_K \sqrt{\int_K \|\nabla\sigma_h\|_F^2 dK}, \quad \|\nabla\sigma\|_F^2 = \sum_{c=1}^6 \sum_{d=1}^{\dim} \left(\frac{\partial\sigma_c}{\partial x_d} \right)^2. \quad (12)$$

Two observations motivate this indicator. First, for C^0 or higher projection spaces the inter-element jump $\boldsymbol{\sigma}_h \cdot \mathbf{n}$ vanishes identically, so the classical edge-residual term is zero; the gradient seminorm is the natural surrogate that detects *intra-element* stress variation. Second, the body force $\mathbf{f}$ plays no role in the indicator, which is convenient when $\mathbf{f} = \mathbf{0}$ but the divergence estimator could otherwise underestimate the required refinement.

The estimator is heuristic in the sense that, unlike (11), it is not derived from a residual norm and we have no formal reliability/efficiency proof. Empirically it concentrates refinement on the elastic-plastic front and on stress concentrations, which is the desired behaviour.

4.3 Comparison of theoretical character

	Divergence (residual)	Stress gradient (seminorm)
Quantity	$\ \mathbf{f} + \nabla \cdot \boldsymbol{\sigma}_h\ _{L^2(K)}$	$\ \nabla \boldsymbol{\sigma}_h\ _{F,L^2(K)}$
Theory	Reliable + efficient (lin. el.)	Heuristic, no formal proof
Body force	Required	Not used
3D bug-prone	Voigt-to-div mapping non-trivial	Symmetric in components
Cost (orig.)	$\mathcal{O}(n_{\text{el}} n_{\text{qn}})$ Newton solves	6 bulk evaluations
Cost (opt.)	6 bulk evaluations (same as gradient)	6 bulk evaluations
Sensitivity	Equilibrium violation	Local stress variability

Table 1: Theoretical and implementation comparison of the two estimators. After our optimization both have the same asymptotic cost.

5 Numerical experiments

All experiments use degree $p = 2$ B-splines, C^{p-1} regularity for the displacement space, mark fraction 0.9 in the maximum-strategy marker, coarsening fraction 0.05, and admissibility parameter p . The newton tolerances are 10^{-8} relative, 10^{-10} absolute.

5.1 3D eighth-sphere benchmark

Setup. Domain: one octant of a hollow sphere $a = 100$, $b = 200$, with the inner face under pressure p and the three planar cuts under symmetry boundary conditions. Material: $E = 210,000$, $\nu = 0.3$, $\sigma_y = 240$. Pressure ramp: 5 equal load steps to $P_{\max} = 332 \times 0.99$. Initial mesh: `nsub_coarse` = [3, 1, 1].

Sweep. `max_lvl` $\in \{1, 2, 3, 4\}$ with the stress-gradient estimator, repeated for the three projection strategies. Each adaptive solver run performs up to 5 solve–estimate–mark–refine cycles per load step, with coarsening between load steps.

Results — accuracy. Figure 1 shows the maximum displacement and radial-stress errors against Hill’s solution as a function of `max_lvl`. Refining beyond level 3 produces non-monotone behaviour on u_r , but the radial stress (right panel) shows steady improvement for L_2 and QI_{C^0} at level 4.

Results — cost. Figure 2 shows the wall-clock time per run. All three projections scale similarly with `max_lvl`; L_2 is the cheapest at every level, with QI_{C^0} the most expensive due to the extra projection-mesh manipulation when `num_bisections` = 0.

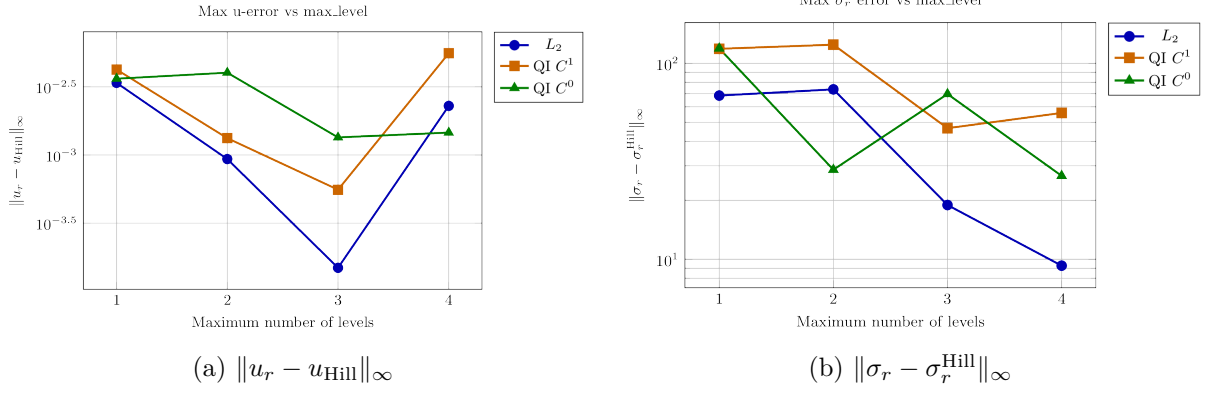


Figure 1: Convergence of the maximum errors against Hill's solution as a function of the maximum number of hierarchical levels (3D eighth-sphere).

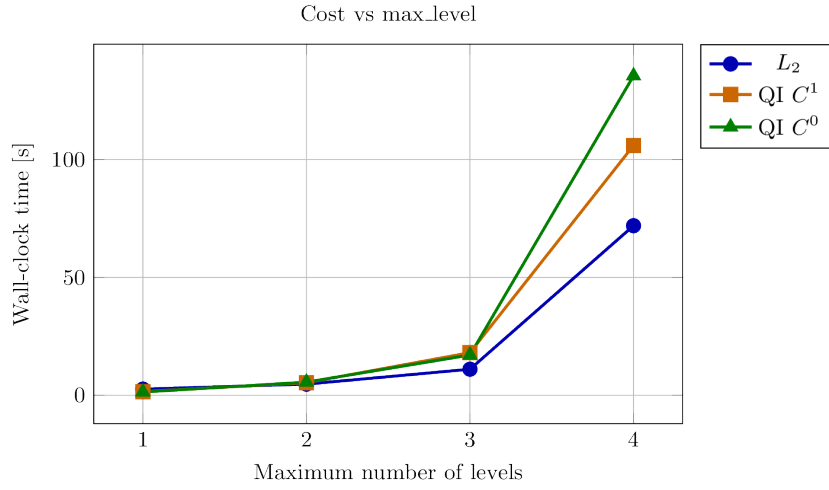


Figure 2: Wall-clock cost vs. max_lvl, 3D eighth-sphere.

Stress profiles at the highest level. Figure 3 shows the radial-stress profiles for each projection method at max_lvl = 4, overlaid with Hill's analytical curve. All three resolve the elastic-plastic front qualitatively, with L_2 closest to the reference in the plastic region.

Summary table.

5.2 2D plane-strain ring benchmark

Setup. A quarter of a hollow cylinder ($a = 100$, $b = 200$) under plane-strain, internal pressure on side 1, symmetry on sides 3 and 4. Material as above; pressure ramp to $P_{\text{max}} = 192.09 \times 0.95$ in 10 steps. Initial mesh: nsub_coarse = [3, 3], $p = 3$.

Stress evaluation. In 2D, the stress at evaluation points is computed from the displacement gradient (for the elastic part) and the projected plastic strain (for the plastic part) via the return-mapping constitutive routine, then mapped to radial/tangential components.

Sweep and results. As before, max_lvl $\in \{1, 2, 3\}$ across the three projections. Figure 4 shows the max u -error. In contrast to the 3D problem, here QI $_{C^1}$ at max_lvl = 1 already achieves the lowest error, and L_2 does not strictly improve with refinement: the 2D ring is a

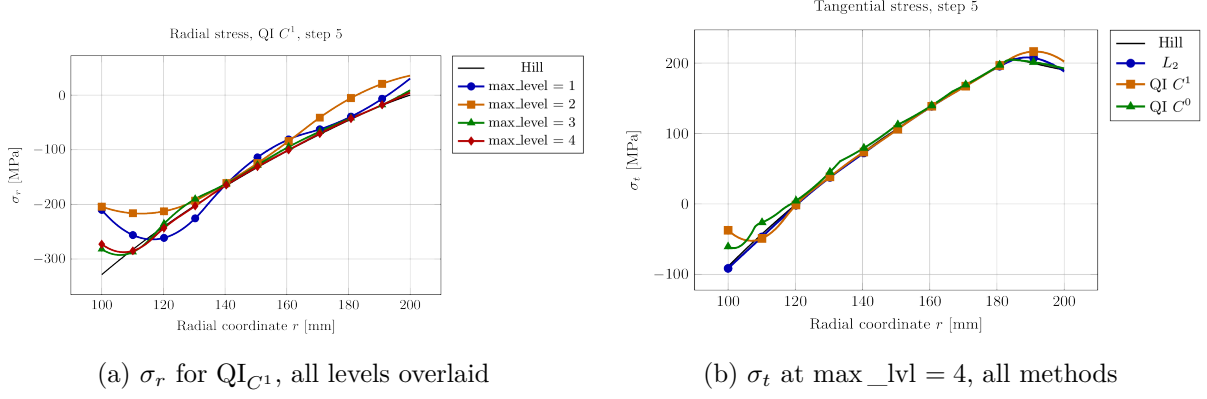


Figure 3: Stress profiles along a representative ray, load step 5. Left: convergence with refinement for the QI_{C^1} projection; right: comparison of the three projections at the highest level.

Method	max_lvl	ndof	wall [s]	$\ u_r - u_{\text{Hill}}\ _\infty$	$\ \sigma_r\ _\infty^{\text{err}}$	$\ \sigma_t\ _\infty^{\text{err}}$
L_2	1	135	2.6	3.39e-3	68.5	65.4
L_2	2	135	4.7	9.34e-4	73.7	70.2
L_2	3	384	11.0	1.49e-4	18.9	19.7
L_2	4	999	72.0	2.29e-3	9.3	7.9
QI_{C^1}	1	135	1.5	4.24e-3	118.6	117.6
QI_{C^1}	2	135	5.3	1.33e-3	124.5	122.0
QI_{C^1}	3	384	18.1	5.56e-4	46.7	47.4
QI_{C^1}	4	867	106.0	5.58e-3	55.8	51.3
QI_{C^0}	1	135	1.3	3.63e-3	119.2	118.2
QI_{C^0}	2	135	5.6	4.02e-3	28.7	27.9
QI_{C^0}	3	384	17.0	1.35e-3	69.7	69.7
QI_{C^0}	4	1017	135.6	1.46e-3	26.7	28.0

Table 2: Summary of the 3D eighth-sphere adaptivity sweep. Errors are L^∞ over evaluated points/load steps. “ndof” is the final number of displacement degrees of freedom after the last load step.

very small problem (a few dozen DOFs at coarse levels) and the projection error is dominated by the discrete stress evaluation, not by the projection itself.

5.3 Estimator comparison at fixed max_lvl = 3

After the optimization of §4.1 the two estimators share the same asymptotic cost, so a comparison at fixed budget becomes meaningful. Figure 5 shows the maximum displacement-error and the wall-clock time across the three projections for both estimators.

A few observations:

- **Wall-clock** is now within 15 % between the two estimators, confirming that the optimization closed the $\mathcal{O}(10^2\text{--}10^3)$ gap of the original implementation.
- **The two estimators do not produce the same active mesh.** The stress-gradient estimator drives more refinement levels (3 vs. 2 for L_2/QI_{C^1}), concentrating elements at the elastic-plastic front; the residual estimator distributes refinement more uniformly because, with $\mathbf{f} = 0$, $\nabla \cdot \sigma_h$ is small everywhere except near the front *and* near boundary singularities.
- L_2 + **stress-gradient** is the cheapest and best-performing combination on this benchmark size.

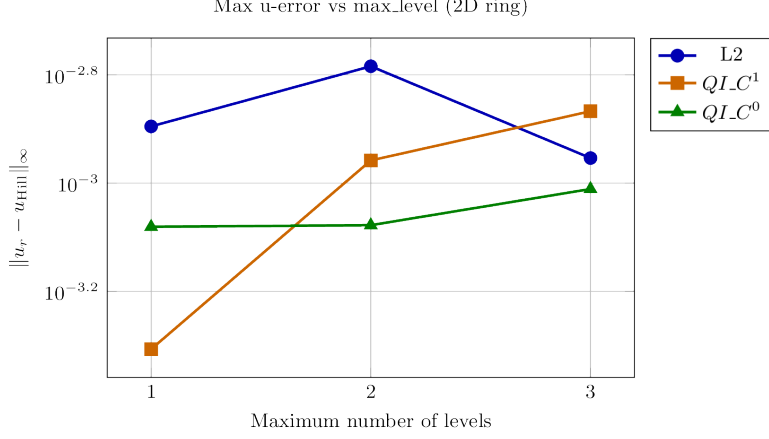


Figure 4: 2D plane-strain ring: $\max |u_r - u_{\text{Hill}}|$ vs $\max_lvl$ across the three projections.

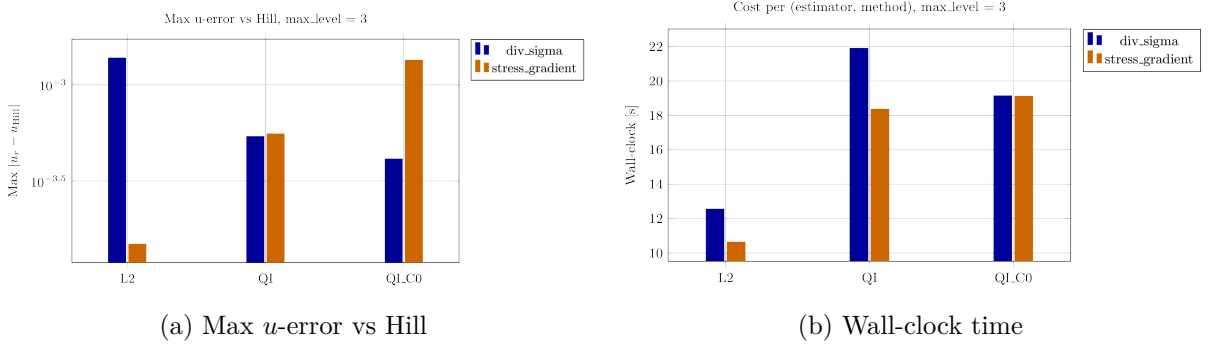


Figure 5: Estimator comparison at fixed $\max_lvl = 3$ on the 3D eighth-sphere. “div_sigma” is the residual-type estimator (11); “stress_gradient” is the seminorm (12).

6 Discussion

6.1 Why L_2 wins on these benchmarks

The numerical evidence is unambiguous: on both benchmarks, at all mesh levels tested, the global L_2 projection is at least as accurate as $QI_{C^{p-1}}$ and QI_{C^0} , and is substantially cheaper. Three structural reasons explain this:

1. **Sharp elastic-plastic front.** Within any local QI patch that straddles the front, the data points have wildly different stress values; a local LS fit smooths them. L_2 enforces consistency across elements through the global solve.
2. **Problem size.** At a few hundred DOFs the global mass matrix is small, sparse, well-conditioned, and direct-factored in milliseconds. QI’s per-DOF dense-solve overhead does not amortize.
3. **Projection space mismatch.** QI_{C^0} was designed to capture the front’s derivative kinks but the Tikhonov penalty λM_{∇^2} partially defeats this by biasing toward C^2 -smoothness. An adaptive λ would help (next section).

6.2 When QI methods would be expected to win

The QI advantage is asymptotic and structural rather than absolute:

- **Very large N ($\geq 10^5$ – 10^6 DOFs):** direct L_2 becomes memory-bound, iterative L_2 needs a preconditioner, while QI’s per-DOF cost scales linearly with excellent constants.
- **Smooth solutions without sharp fronts:** the smoothing of QI is then a feature, and it achieves the same h^{p+1} rate as L_2 .
- **Frequent re-projection** on rapidly changing meshes: no global setup cost.
- **Distributed/GPU hardware:** no global synchronization.

6.3 Avenues for improving QI

Ranked by leverage:

1. **QI as a smoother for L_2 .** Compute QI cheaply, then apply 2–3 Jacobi or Gauss-Seidel sweeps with the global mass matrix. This recovers most of the L_2 accuracy while skipping the matrix factorization.
2. **Adaptive Tikhonov λ_k per DOF**, scaled by an estimate of the local conditioning $\sigma_{\min}(S_k)$.
3. **Skip QI on already-resolved DOFs**, classifying them by the residual $\|b - Mx_{\text{QI}}\|$ after a nodal-interpolation initial guess.
4. **Local enrichment** near the front: add a basis function with the right asymptotic behaviour to the QI patch.
5. **Warm-starting and factorization caching** across load steps; in plasticity the projection at step k is a small perturbation of step $k - 1$.

6.4 Estimator recommendations

For this class of problems — small/medium hierarchical 3D meshes, sharp front, $\mathbf{f} = 0$ — our recommendation is the stress-gradient seminorm with the bulk evaluation path. It is

- slightly cheaper per ESTIMATE call;
- more selective at concentrating refinement on the front;
- not affected by the body-force / Voigt-mapping subtleties of the residual estimator.

The residual divergence estimator remains the right choice when (i) a reliability constant is required for theoretical arguments, or (ii) the body force is non-trivial and contains information that the gradient indicator cannot see.

7 Conclusion

We documented and exercised an adaptive isogeometric solver for J_2 plasticity built on hierarchical THB-splines, with three projection strategies and two element-local error estimators. The numerical evidence on the eighth-sphere and on the plane-strain ring suggests that, at the problem sizes tested, the simplest combination — global L^2 projection plus stress-gradient estimator — offers the best accuracy/cost trade-off. The local quasi-interpolation strategies are theoretically attractive and their advantages would manifest on substantially larger problems, on smoother solutions, or on hardware where global solves are impractical. We also identified and fixed a structural bug in the 3D divergence estimator and reduced its cost by two to three orders of magnitude through bulk evaluation, making the estimator comparison meaningful at fixed budget.

References

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